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$$I = \int_1^{+\infty} x^{-\frac{2}{3}} dx$$

$$\int x^{-\frac{2}{3}} dx = \frac{x^{\frac{1}{3}}}{\frac{1}{3}} = 3x^{\frac{1}{3}} + C = 3\sqrt[3]{x} + C$$

$$I = \lim_{b \rightarrow +\infty} [3\sqrt[3]{x}]_1^b = \lim_{b \rightarrow +\infty} (3\sqrt[3]{b} - 3) = +\infty \Rightarrow \text{Divergent}$$

References